

Robust Nash Preference Optimization: Aligning Language Models under Heterogeneous Preferences

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Abstract

Aligning language models increasingly relies on multiple, conflicting preference signals: helpfulness, harmlessness, conciseness, and domain-specific judges routinely disagree. Standard practice scalarizes these signals into a fixed average, which can hide a severe failure on a minority objective, while assigning each objective its own player yields a general-sum game for which last-iterate convergence guarantees are unavailable. We propose Robust Nash Preference Optimization (RONPO), which keeps heterogeneous alignment inside a two-player game: a single adversary jointly selects a preference oracle and an opponent response, and the policy must remain competitive against the resulting worst-case pair. The induced objective is a KL-regularized bilinear saddle whose inner problem is a distributionally robust soft minimum over objective–response pairs. We prove that the game admits a unique robust Nash policy, that exact mirror-descent dynamics converge to it at an $O(1/T)$ last-iterate rate, and that the same rate holds in expectation under single-pair Bernoulli preference feedback. The exact update reduces to a partition-free log-ratio regression against a sampled adversarial pair, giving a practical loss for language models. On synthetic preference games, RONPO nearly closes the gap to the robust linear-programming optimum where averaging-based methods collapse on a hidden objective, and its dynamics track the predicted convergence rate. At model scale, RONPO improves worst-objective reward and is preferred by an external judge over averaged and single-oracle baselines.

Introduction

Reinforcement learning from human feedback (RLHF) is the standard recipe for aligning large language models: pairwise human preferences are compressed into a scalar reward model, and the policy is optimized against that reward under a KL constraint (Christiano et al. 2017; Ouyang et al. 2022). Direct alignment methods such as DPO shortcut the reward-modeling step but retain the same scalar, Bradley–Terry structure (Rafailov et al. 2023; Bradley and Terry 1952). This compression is a modeling assumption, not a fact about human judgment: a Bradley–Terry model induces a transitive order over responses, whereas aggregate human pref-

erences can be cyclic or context-dependent (Tversky 1969; May 1954).

Why a game? When preferences need not be transitive, there may exist no scalar utility whose maximizer is “the preferred response,” so reward maximization is no longer the right solution concept. Nash learning from human feedback (NLHF) instead works with a general preference oracle $P(y \succ y' \mid x)$ and seeks a policy that no opponent policy can beat in expectation—a von Neumann winner of the induced two-player constant-sum game (Munos et al. 2024; Dudík et al. 2015; Azar et al. 2024). The game-theoretic view is what supplies robustness: the learned policy is evaluated against an explicit opponent that actively searches for responses on which the policy loses, rather than against a fixed scalar score. Self-play methods such as SPPO and INPO make this practical for language models via no-regret mirror descent, and INPO in particular provides *last-iterate* convergence to the regularized Nash policy (Wu et al. 2025; Zhang et al. 2025).

The heterogeneity problem. Realistic alignment, however, rarely involves a single preference source. A model is judged simultaneously by helpfulness, harmlessness, factuality, conciseness, style, and domain-policy critics, and these judges conflict. Existing methods handle this in one of two ways, and both are problematic. (i) *Averaging.* SPPO- and INPO-style methods, like DPO and its variants, consume one oracle; under K judges they are applied to a fixed weighted average $\bar{P} = \sum_k w_k P_k$. A policy can then score highly on average while failing catastrophically under a minority judge—precisely the failure a safety or truthfulness critic is supposed to prevent, and one we isolate experimentally. (ii) *One player per objective.* Heterogeneous multi-player extensions such as MNPO assign each oracle to its own player (Wu et al. 2026). This is algorithmically natural but structurally costly: with different oracles the game is general-sum, the bilinear cross terms no longer cancel, and the monotone variational-inequality machinery behind last-iterate convergence is unavailable. Scalar robust methods such as MaxMin-RLHF (Chakraborty et al. 2024) and group-robust preference optimization (Ramesh et al. 2024) restore a worst-case flavor, but they operate on scalar rewards or group losses and their adversary ranges only over objective *weights*, not over the opponent responses that a preference

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game is built to withstand.

This work. RONPO keeps the heterogeneity but changes where it lives. There is one policy player and one adversary, and the adversary’s action is a *pair* (k, a) : which preference oracle judges the policy, and which opponent response the policy must beat. Confining heterogeneity to a single adversary’s action set keeps the problem a two-player KL-regularized bilinear saddle, so the strongly monotone structure—and with it existence, uniqueness, and last-iterate convergence—survives arbitrary conflict among the K oracles. The inner minimization is exactly the dual of a Kullback–Leibler distributionally robust program over pairs: RONPO maximizes a regularized *worst-case floor* on win probability, with a single temperature κ interpolating between hard worst-case and plain averaging.

Our contributions are as follows.

- **Formulation.** We cast heterogeneous preference alignment as an objective-adversarial game whose adversary selects objective–response pairs. The inner problem is a KL-DRO soft minimum (Prop. 1), and the pair-level floor certifies strictly more than weight-only max-min robustness (Remark 1).
- **Guarantees.** The game has a unique robust Nash policy (Thm. 1); exact mirror-descent dynamics converge to it at an $O(1/T)$ last-iterate rate (Thm. 2); and the same rate holds in expectation under single-pair Bernoulli preference feedback (Cor. 1). To our knowledge, these are the first last-iterate guarantees for general-preference alignment with heterogeneous objectives; per-player heterogeneous formulations fall outside the monotone class, so no analogous guarantee is available there.
- **Algorithm.** The exact update reduces to a partition-free log-ratio regression against a sampled adversarial pair (Prop. 2), yielding an INPO-style loss with no partition function. We state precisely which properties the deployed plug-in surrogate inherits from the idealized loss and which it does not.
- **Experiments.** Synthetic preference games isolate the hidden-objective failure of averaging—RONPO nearly closes the gap to the robust LP optimum at essentially no average cost—and numerically validate the convergence theory. At model scale, RONPO improves average and worst-objective reward over averaged and single-oracle baselines and is preferred by an external judge.

We are explicit about scope: the equilibrium and rate analysis instantiates standard strongly monotone variational-inequality arguments (Facchinei and Pang 2003; Nemirovski and Yudin 1983); the contribution is the formulation that makes those arguments *apply* under heterogeneity, together with the resulting algorithm and guarantees.

Related Work

RLHF and direct alignment. Reward-based RLHF optimizes a KL-regularized policy against a learned scalar reward (Christiano et al. 2017; Ouyang et al. 2022). DPO removes the explicit reward model through a closed-form reparameterization (Rafailov et al. 2023); IPO replaces its

logistic link with a squared loss on preference probabilities to reduce overfitting (Azar et al. 2024); KTO uses pointwise, prospect-theoretic feedback (Ethayarajh et al. 2024); and SimPO drops the reference model (Meng, Xia, and Chen 2024). All inherit a single, implicitly transitive preference signal; under heterogeneous judges they are applied to an averaged or single oracle, the regime our synthetic study shows can mask minority-objective collapse.

General preferences and Nash alignment. Dueling and contextual dueling bandits introduced learning from pairwise comparisons and the von Neumann winner solution concept (Yue et al. 2012; Dudík et al. 2015). NLHF brought this view to LLM alignment (Munos et al. 2024), with self-play and no-regret instantiations including SPPO, INPO, DNO, SPO, and online IPO variants (Wu et al. 2025; Zhang et al. 2025; Rosset et al. 2024; Swamy et al. 2024; Calandriello et al. 2024), and theoretical treatments of the KL-regularized preference game (Ye et al. 2024). MNPO extends NLHF to population opponents and, in its heterogeneous variant, to per-player oracles (Wu et al. 2026). RONPO reuses the mirror-descent foundation but selects heterogeneous objectives adversarially rather than averaging them, and—unlike the per-player split—retains the two-player monotone structure (Lemma 1).

Robust and multi-objective alignment. MaxMin-RLHF optimizes an egalitarian max-min over scalar reward models (Chakraborty et al. 2024); group-robust preference optimization minimizes a worst-group DPO-style loss (Ramesh et al. 2024); MODPO folds multiple objectives into a single direct loss (Zhou et al. 2024); rewarded soups and directional alignment trace Pareto fronts by weight interpolation (Ramé et al. 2023; Wang et al. 2024). RONPO differs on two axes: it works with general preference *oracles* rather than scalar rewards, retaining the intransitivity that motivates NLHF, and its adversary ranges over response pairs a in addition to objective indices k , so it can expose failures that depend jointly on the judge and the opponent response (Remark 1). Its inner problem is a KL-constrained DRO value—an entropic risk (Hu and Hong 2013)—with the specific feature that the ambiguity set is coupled to a policy player through a bilinear matrix, which is what keeps the saddle monotone.

Preliminaries

Preference-Based Alignment as a Contextual Bandit

We work in the preference-based contextual bandit setting that underlies RLHF (Yue et al. 2012; Dudík et al. 2015). A context (prompt) $x \in \mathcal{X}$ is drawn from a distribution ρ ; the learner selects a response (action) y from a finite response set $\mathcal{Y}$; and feedback is *relative*: for a pair of responses (y, y') , a preference oracle returns a binary label

$$z \sim \text{Ber}(P(y \succ y' \mid x)), \quad z \in \{0, 1\}, \quad (1)$$

where $z = 1$ means y is preferred to y' . The oracle $P : \mathcal{Y} \times \mathcal{Y} \times \mathcal{X} \rightarrow [0, 1]$ is skew-symmetric, $P(y \succ y' \mid x) + P(y' \succ y \mid x) = 1$ and $P(y \succ y \mid x) = 1/2$, but need not be induced by any scalar utility: cyclic preferences are allowed. A policy maps each context to a distribution $\pi(\cdot \mid$

$x) \in \Delta(\mathcal{Y})$, and for policies π, π' we write $P(\pi \succ \pi' \mid x) = \mathbb{E}_{y \sim \pi(\cdot \mid x), y' \sim \pi'(\cdot \mid x)}[P(y \succ y' \mid x)]$.

All objectives in this paper are expectations over $x \sim \rho$ of per-context terms, and the exact dynamics we analyze decompose across contexts. Following standard practice in NLHF analysis (Munos et al. 2024; Wu et al. 2025; Zhang et al. 2025), we therefore state definitions and results for a fixed context and drop x from the notation; every statement lifts to the contextual case by conditioning on x and averaging over $x \sim \rho$. The parametric algorithm (Alg. 1) trains a single prompt-conditioned network across contexts; the gap between the tabular guarantee and the parametric trainer is discussed under Limitations.

Nash Learning from General Preferences

With the context fixed, a policy is a probability vector $\pi \in \Delta(\mathcal{Y})$. The unregularized NLHF objective is $\max_{\pi} \min_{\pi'} P(\pi \succ \pi')$, whose solution is a von Neumann winner: a policy that no opponent policy beats in expectation (Munos et al. 2024; Dudík et al. 2015). KL regularization toward a fixed reference policy $\mu \in \text{int } \Delta(\mathcal{Y})$ (the role played by π_{ref} in direct alignment) yields the smoother saddle objective

$$J(\pi, \pi') = P(\pi \succ \pi') - \tau \text{KL}(\pi \parallel \mu) + \tau \text{KL}(\pi' \parallel \mu), \quad (2)$$

with the first player maximizing and the second minimizing, and $\tau > 0$ the regularization temperature.

No-Regret Learning via Mirror Descent

A two-player saddle game is two coupled online convex problems, and if both players run no-regret algorithms, the duality gap of average play vanishes at rate $(R_T^\pi + R_T^{\pi'})/T$ (Freund and Schapire 1999; Cesa-Bianchi and Lugosi 2006); Appendix B reviews this reduction. The standard no-regret rule on the simplex is entropic online mirror descent (OMD): for losses with subgradients g_t at the iterate u_t ,

$$u_{t+1} = \arg \min_{v \in \Delta(\mathcal{Y})} \{ \eta_t \langle g_t, v \rangle + \text{KL}(v \parallel u_t) \}, \quad (3)$$

whose closed form is the multiplicative-weights update

$$u_{t+1}(y) = \frac{u_t(y) \exp(-\eta_t g_t(y))}{\sum_a u_t(a) \exp(-\eta_t g_t(a))}, \quad (4)$$

with regret $R_T \leq G \sqrt{2T \log |\mathcal{Y}|}$ for $\|g_t\|_\infty \leq G$ and tuned constant step (Appendix B.2). Average-iterate convergence follows from regret alone; the last-iterate results below additionally exploit strong monotonicity of the regularized operator.

Existing Methods Through the OMD Lens

The self-play baselines instantiate template (4) and differ only in the opponent. **SPPO** (Wu et al. 2025) uses the pointwise self-play update $\pi_{t+1}(y) \propto \pi_t(y) \exp\{\eta(P(y \succ \pi_t) - \frac{1}{2})\}$ with $P(y \succ \pi_t) = \mathbb{E}_{a \sim \pi_t}[P(y \succ a)]$; under heterogeneous preferences it consumes an averaged oracle $\bar{P} = \sum_k w_k P_k$, so a response that is excellent on most objectives but unacceptable on one can still earn a high score.

INPO (Zhang et al. 2025) applies OMD to the regularized game (2),

$$\pi_{t+1}(y) \propto \exp\{\eta P(y \succ \pi_t)\} \mu(y)^{\eta\tau} \pi_t(y)^{1-\eta\tau}, \quad (5)$$

and turns it into a partition-free loss by regressing pairwise log-ratios $\Delta_\pi(y, y') := \log \pi(y) - \log \pi(y')$, a device we reuse. **MNPO** (Wu et al. 2026) replaces the single opponent with a population via a geometric opponent mixture. Its heterogeneous extension assigns oracle P_i to player i ; the game is then neither symmetric nor constant-sum, the bilinear cross terms no longer cancel pairwise, and the operator need not be monotone. We do not claim such dynamics diverge—population play can perform well empirically (Wu et al. 2026; Ramé et al. 2023)—only that the monotone variational-inequality route to last-iterate convergence is unavailable. Table 1 summarizes the comparison; RONPO recovers the monotone structure exactly (Lemma 1) by keeping a single adversary.

Robust Nash Preference Optimization

The Objective-Adversarial Game

Assume K heterogeneous preference oracles $P_1, \dots, P_K$ indexed by $\mathcal{K} = \{1, \dots, K\}$, and let $\mathcal{A} \subseteq \mathcal{Y}$ be a finite adversarial response pool. In LLM applications, $\mathcal{A}(x)$ is constructed from reference responses, current-policy responses, historical checkpoints, external teachers, or best-of- n candidates. The policy chooses $\pi \in \Delta(\mathcal{Y})$; the adversary chooses a distribution $s \in \Delta(\mathcal{K} \times \mathcal{A})$ over objective–response pairs (k, a) . Because $\mathcal{A}$ is finite, the robustness RONPO certifies is relative to this pool and to the K oracles: the learned policy is unexploitable by pairs in $\mathcal{K} \times \mathcal{A}$, not by every response in $\mathcal{Y}$. Enlarging $\mathcal{A}$ strengthens the guarantee at higher cost; we treat pool construction as an empirical design choice.

Define the win functional

$$W(\pi, s) = \mathbb{E}_{y \sim \pi, (k, a) \sim s}[P_k(y \succ a)]. \quad (6)$$

Let $\mu \in \text{int } \Delta(\mathcal{Y})$ be the reference policy and $s_0 \in \text{int } \Delta(\mathcal{K} \times \mathcal{A})$ an adversary prior. RONPO solves

$$\begin{aligned} \max_{\pi \in \Delta(\mathcal{Y})} \min_{s \in \Delta(\mathcal{K} \times \mathcal{A})} V(\pi, s) &:= W(\pi, s) - \tau \text{KL}(\pi \parallel \mu) \\ &\quad + \kappa \text{KL}(s \parallel s_0), \end{aligned} \quad (7)$$

with temperatures $\tau, \kappa > 0$. Writing $A \in \mathbb{R}^{|\mathcal{Y}| \times K|\mathcal{A}|}$ for the matrix $A_{y, (k, a)} = P_k(y \succ a)$, we have $W(\pi, s) = \pi^\top A s$, and the saddle operator is

$$\begin{aligned} F(\pi, s) &= (-\nabla_\pi V(\pi, s), \nabla_s V(\pi, s)) \\ &= (-A s + \tau(\log(\pi/\mu) + \mathbf{1}), \\ &\quad A^\top \pi + \kappa(\log(s/s_0) + \mathbf{1})), \end{aligned} \quad (8)$$

with logarithms and divisions componentwise. If $K = 1$, $\mathcal{A} = \mathcal{Y}$, $s_0 = \mu$, and $\kappa = \tau$, then (7) reduces to the KL-regularized NLHF game (2), so RONPO strictly generalizes single-oracle NLHF.

Method	Opponent object	Oracle choice	Pointwise win rate?	Heterogeneity	Last-iterate
DPO / IPO / KTO / SimPO	offline pair data	fixed (implicit)	—	No	—
SPPO	current policy	fixed	yes	No	Yes
INPO	current policy + reference	fixed	no	No	Yes
MNPO	opponent population	shared or mixed	no	Yes	No [†]
RONPO (ours)	objective–response pair	adversarial	no	Yes	Yes

Table 1: Modeling comparison. The crucial distinction is the opponent object. “Heterogeneity” marks whether the method natively represents conflicting objectives without averaging; “Last-iterate” marks whether a last-iterate convergence guarantee to the (regularized) Nash policy is available. Direct-alignment methods optimize a fixed offline objective and are listed for completeness. [†]The heterogeneous (per-player oracle) MNPO variant is general-sum and falls outside the monotone class, so the homogeneous last-iterate argument does not transfer.

A Distributionally Robust Interpretation

The adversary’s KL term makes (7) a regularized worst case rather than a hard one, and the inner problem has a closed form.

Proposition 1 (Soft worst-case form). *For fixed π , let $c_\pi(k, a) = \mathbb{E}_{y \sim \pi}[P_k(y \succ a)]$. Then*

$$\min_s V(\pi, s) = -\kappa \log \mathbb{E}_{(k,a) \sim s_0} \left[\exp \left(-\frac{c_\pi(k, a)}{\kappa} \right) \right] - \tau \text{KL}(\pi \| \mu), \quad (9)$$

and as $\kappa \downarrow 0$ the first term converges to $\min_{k,a} c_\pi(k, a)$.

The proof (Appendix A) is the Gibbs variational principle. Equation (9) is exactly what RONPO protects: a regularized soft minimum over objective–response pairs—a worst-case floor on win probability. The penalized inner problem is the Lagrangian form of a KL-DRO objective over pair distributions centered at s_0 (Hu and Hong 2013), and κ is the single robustness knob: large κ keeps s near the prior and recovers fixed scalarization; small κ approaches a hard selector of the currently weakest pair. We use the soft minimum rather than a hard constraint $c_\pi(k, a) \geq \rho$ because a constraint requires a floor ρ that is known and feasible a priori (rarely available per judge), whereas the soft minimum raises the floor as far as the geometry allows, keeps the problem an unconstrained saddle so the monotone machinery applies, and exposes the interpolation above.

Remark 1 (Pair-level vs. weight-only robustness). *For any fixed opponent distribution $q \in \Delta(\mathcal{A})$, the weight-only floor optimized by scalar max-min methods (Chakraborty et al. 2024) satisfies $\min_{k \in \mathcal{K}} \mathbb{E}_{a \sim q}[c_\pi(k, a)] \geq \min_{(k,a) \in \mathcal{K} \times \mathcal{A}} c_\pi(k, a)$ for every policy π (Appendix A). Hence any level certified on RONPO’s pair floor is simultaneously certified on every weight-only floor with opponents in $\mathcal{A}$, while the converse fails: a policy can pass all weight-only floors yet lose badly to one specific judge–response pair, a failure mode only the pair-level adversary exposes.*

Equilibrium Guarantees

The reason to confine heterogeneity to one adversary is structural: the preference coupling then cancels from the monotonicity gap, no matter how much the oracles conflict.

Lemma 1 (Coupling cancels in the monotonicity gap). *For any interior $z = (\pi, s)$ and $z' = (\pi', s')$,*

$$\begin{aligned} \langle F(z) - F(z'), z - z' \rangle &= \tau \{ \text{KL}(\pi \| \pi') + \text{KL}(\pi' \| \pi) \} \\ &\quad + \kappa \{ \text{KL}(s \| s') + \text{KL}(s' \| s) \} \\ &\geq m \{ D(z \| z') + D(z' \| z) \}, \end{aligned} \quad (10)$$

where $m = \min\{\tau, \kappa\}$ and $D(z \| z') = \text{KL}(\pi \| \pi') + \text{KL}(s \| s')$.

Proof idea. The bilinear block $(-As, A^\top \pi)$ is skew-symmetric, so its contribution $-\langle A\delta s, \delta \pi \rangle + \langle A^\top \delta \pi, \delta s \rangle$ vanishes identically—for every matrix A , hence for every collection of oracles and every pool. The separable entropic regularizers contribute the symmetrized KL terms. Full details are in Appendix A. $\square$

Theorem 1 (Existence and uniqueness). *The game (7) has a unique saddle point $z^* = (\pi^*, s^*)$, and both components have full support.*

Existence follows from Brouwer’s theorem applied to the continuous Gibbs best-response maps; uniqueness combines the saddle variational inequality with the strong monotonicity of Lemma 1 (Appendix A). We call π^* the *robust Nash policy*.

Last-Iterate Convergence

We instantiate OMD (3) on the product $Z = \Delta(\mathcal{Y}) \times \Delta(\mathcal{K} \times \mathcal{A})$ with proximal term $D(z \| z_t) = \text{KL}(\pi \| \pi_t) + \text{KL}(s \| s_t)$; since both the feasible set and the regularizer decompose over blocks, one product step equals two per-block steps,

$$z_{t+1} = \arg \min_{z \in Z} \{ \eta_t \langle F(z_t), z \rangle + D(z \| z_t) \}. \quad (11)$$

For $z_t = (\pi_t, s_t)$, define the policy-side adversarial reward and adversary-side cost

$$r_t(y) = \mathbb{E}_{(k,a) \sim s_t} [P_k(y \succ a)] = (As_t)(y), \quad (12)$$

$$c_t(k, a) = \mathbb{E}_{y \sim \pi_t} [P_k(y \succ a)] = (A^\top \pi_t)(k, a). \quad (13)$$

Substituting (8) into (4) gives the closed-form updates

$$\pi_{t+1}(y) \propto \exp\{\eta_t r_t(y)\} \mu(y)^{\eta_t \tau} \pi_t(y)^{1-\eta_t \tau}, \quad (14)$$

$$s_{t+1}(k, a) \propto \exp\{-\eta_t c_t(k, a)\} s_0(k, a)^{\eta_t \kappa} s_t(k, a)^{1-\eta_t \kappa}. \quad (15)$$

The policy upweights responses that beat the current adversarial mixture; the adversary upweights pairs on which the current policy is weakest. Let $\Theta_t = D(z^* \| z_t)$.

Lemma 2 (Verifiable operator bound). *Initialize $z_0 = (\mu, s_0)$ and suppose $\eta_t \tau \leq 1$ and $\eta_t \kappa \leq 1$ for all t . Then the iterates of (14)–(15) satisfy $\|F(z_t)\|_{\oplus,*} \leq G$ with $G = 2 + \max\{\tau, \kappa\}$, where $\|g\|_{\oplus,*} = \max\{\|g_\pi\|_\infty, \|g_s\|_\infty\}$.*

Theorem 2 (Exact last-iterate rate). *Under the conditions of Lemma 2, the exact dynamics satisfy*

$$\Theta_{t+1} \leq (1 - m\eta_t)\Theta_t + \eta_t^2 G^2, \quad (16)$$

with $m = \min\{\tau, \kappa\}$. With $\eta_t = 2/(m(t + t_0))$ and t_0 large enough that $\eta_t \tau \leq 1$ and $\eta_t \kappa \leq 1$, the last iterate obeys $\Theta_T \leq M/(T + t_0)$ with $M = \max\{t_0\Theta_0, 4G^2/m^2\}$, i.e., $\Theta_T = O(G^2/(m^2T))$.

Proof idea. Add the one-step entropic-OMD descent inequality over the two blocks with comparators (π^*, s^*) ; the saddle variational inequality makes $\langle F(z^*), z_t - z^* \rangle \geq 0$ and Lemma 1 lower-bounds the remainder by $m\Theta_t$; solve the scalar recurrence (Appendix A). $\square$

The practical estimator (next section) queries preferences stochastically, so we also record that the rate survives Bernoulli feedback: at round t , sample one pair $(k_t, a_t) \sim s_t$ and set $\hat{r}_t(y) = \xi_t(y)$ with $\xi_t(y) \sim \text{Ber}(P_{k_t}(y \succ a_t))$ for each y ; independently sample $y_t \sim \pi_t$ and set $\hat{c}_t(k, a) = \zeta_t(k, a)$ with $\zeta_t(k, a) \sim \text{Ber}(P_k(y_t \succ a))$; run (14)–(15) with $(\hat{r}_t, \hat{c}_t)$ in place of (r_t, c_t) .

Corollary 1 (Stochastic preference feedback). *Under the step-size conditions of Theorem 2, the stochastic dynamics above satisfy $\mathbb{E}[\Theta_T] \leq M/(T + t_0)$ with the same constants: $\hat{r}_t, \hat{c}_t \in [0, 1]$ coordinatewise, so the bound of Lemma 2 holds pathwise, and $\mathbb{E}[(\hat{r}_t, \hat{c}_t) \mid \mathcal{F}_t] = (r_t, c_t)$, so the recurrence (16) holds in conditional expectation (Appendix A).*

These rates certify the idealized tabular dynamics that the practical loss regresses toward; they are not claims about the parametric prompt-conditioned trainer (see Limitations), matching how guarantees are scoped in prior NLHF practice (Zhang et al. 2025).

From Exact Updates to a Practical Loss Partition-Free Log-Ratio Regression

Estimating $r_t(y)$ for every response is expensive, and the normalizer in (14) is inconvenient for parametric policies. Both problems disappear in pairwise log-ratios. For a full-support u , let $\Delta_u(y, y') := \log u(y) - \log u(y')$; all exact iterates are full-support because π_0, μ, s_0 are. Taking log-differences of (14) for two responses cancels the normalizer and yields, with

$$h_t(\pi; y, y') := \Delta_\pi(y, y') - \eta_t \tau \Delta_\mu(y, y') - (1 - \eta_t \tau) \Delta_{\pi_t}(y, y'), \quad (17)$$

the exact-update identity

$$h_t(\pi_{t+1}; y, y') = \eta_t (r_t(y) - r_t(y')). \quad (18)$$

The left side contains only log-probability differences—directly computable for an LLM policy—and the unknown partition function is gone. The right side is estimated without integrating over s_t : sample one pair $(k, a) \sim s_t$, query the oracle twice, $z_y \sim \text{Ber}(P_k(y \succ a))$ and $z_{y'} \sim \text{Ber}(P_k(y' \succ a))$, and use the signed difference $z_y - z_{y'}$, which is conditionally unbiased for $r_t(y) - r_t(y')$. The idealized population loss is

$$L_t(\pi) = \mathbb{E}[(h_t(\pi; y, y') - \eta_t(z_y - z_{y'}))^2], \quad (19)$$

with expectation over $y, y' \sim \pi_t$ and $(k, a) \sim s_t$.

Proposition 2 (Unbiased partition-free regression). *If the policy class contains the exact update (14), then π_{t+1} minimizes (19); if π_t has full support and policies are identified by pairwise log-ratios on that support, the minimizer is unique (Appendix A).*

The Deployed Surrogate

The trainer keeps the log-ratio left-hand side of (18) and makes two substitutions. *First*, oracle queries are replaced by a deterministic plug-in target built from reward-model scores: per-prompt, per-objective scores are min-max normalized over the candidate pool to $\tilde{r}_k(y \mid x)$, and

$$\hat{P}_k(y \succ a \mid x) = \text{sigmoid}(\lambda(\tilde{r}_k(y \mid x) - \tilde{r}_k(a \mid x))), \quad (20)$$

with a fixed preference scale λ . Two distinct policy responses y, y' are drawn from the empirical pool policy and paired with the signed relative target

$$\hat{z}_{k,a}(y, y' \mid x) = \hat{P}_k(y \succ a \mid x) - \hat{P}_k(y' \succ a \mid x), \quad (21)$$

so the fields need not be preference-ordered. *Second*, pairs are selected deterministically from the largest adversary masses of (15) rather than sampled, each weighted by its normalized mass $\bar{w}_{k,a}$ (minibatch mean one). The trained loss is

$$\hat{L}_t(\pi) = \mathbb{E}[\bar{w}_{k,a}(h_t(\pi; y, y') - \eta_t \hat{z}_{k,a}(y, y' \mid x))^2], \quad (22)$$

with Δ_{π_t} precomputed under the policy that generated the pool. Algorithm 1 summarizes the procedure.

Remark 2 (What the surrogate inherits). *Two properties of Prop. 2 survive the substitutions and two do not. Survive: the regression remains partition-free, and $\hat{z}_{k,a}$ is the exact relative preference of the plug-in oracle on the selected pair, so the weighted regression still points toward the OMD step of the surrogate oracle under the reweighted pair distribution. Do not survive: top-mass selection makes the effective adversary a finite-step, mass-truncated approximation of (15), and $\hat{P}_k$ carries plug-in bias governed by λ . Moreover, because (20) is logistic in a scalar score gap, the plug-in oracle is Bradley–Terry and hence transitive: at model scale RONPO exercises heterogeneous-reward robustness, and genuine intransitivity is exercised only in the synthetic games, whose oracles are arbitrary skew-symmetric matrices. We phrase all model-scale claims accordingly.*

Experiments

Our evaluation has two parts. Synthetic preference games (i) isolate the hidden-objective failure of averaging in a setting where the robust optimum is computable exactly, and (ii) numerically validate Theorems 1–2, Corollary 1, and Proposition 1. LLM experiments then test whether the deployed surrogate (Sec.) delivers heterogeneous-reward robustness at model scale.

Algorithm 1 RONPO for prompt-conditioned policies

- 1: Initialize $\pi_0(\cdot | x) = \mu(\cdot | x)$ and adversary prior $s_0(\cdot | x)$ over $\mathcal{K} \times \mathcal{A}(x)$.
- 2: **for** $t = 0, 1, \dots, T - 1$ **do**
- 3: Build or refresh the response pool $\mathcal{A}_t(x)$ (reference, current, and historical policies; external teachers).
- 4: Score $\mathcal{A}_t(x)$ with all K objectives; normalize per prompt-objective; form $\hat{P}_k$ by (20).
- 5: Estimate $\hat{c}_t(k, a | x) = \mathbb{E}_{y \sim \hat{\pi}_t} [\hat{P}_k(y \succ a | x)]$ over the pool.
- 6: Update the adversary by (15); select the largest-mass pairs (k, a) with weights $\bar{w}_{k,a}$.
- 7: For each selected pair, sample distinct $y, y' \sim \hat{\pi}_t(\cdot | x)$; set target $\hat{z}_{k,a}$ by (21); skip near-ties.
- 8: Precompute $\Delta_\mu(y, y')$ and $\Delta_{\pi_t}(y, y')$; update the policy by minimizing $\hat{L}_t$ in (22).
- 9: **end for**
- 10: **return** π_T .

Method	$V_{\min}$	Gap	V_{avg}	Ent.	Decoy
SPPO-avg	0.060	0.269	0.535	0.005	1.000
INPO-avg	0.210	0.119	0.530	1.806	0.290
MNPO-hist-avg	<u>0.213</u>	<u>0.116</u>	0.512	2.043	0.157
HT-MNPO-mix	0.209	0.120	0.530	1.811	0.278
RONPO	0.288	0.041	<u>0.533</u>	1.386	0.340

Table 2: Final metrics in the decoy game ($V^* = 0.3289$). “Gap” is $V^* - V_{\min}$. RONPO attains the best worst-case win rate while preserving a competitive average.

Numerical Validation on Synthetic Games

Setup. There are $|\mathcal{Y}| = 8$ actions and $K = 3$ objectives, each an arbitrary skew-symmetric preference matrix. A weak random tournament background assigns each ordered pair probability 0.56 or 0.44; objective k has a specialist action a_k beating others with probability 0.82; and a *decoy* action beats others with probability 0.78 under objectives 1 and 2 but only 0.06 under objective 3. The reference policy is uniform. We compare five idealized tabular methods: SPPO-avg, INPO-avg, and MNPO-hist-avg on the averaged oracle $\bar{P} = \frac{1}{3} \sum_k P_k$; HT-MNPO-mix, which trains objective-specific players and deploys their arithmetic mixture; and exact RONPO over all objective-action pairs. The central metric is the robust minimum win rate $V_{\min}(\pi) = \min_{k,a} c_\pi(k, a)$, compared against the robust LP benchmark $V^* = \max_\pi V_{\min}(\pi) = 0.3289$ on this instance; we also report the average reference win rate V_{avg} , final entropy, and decoy mass. Construction details and hyperparameters are in Appendix C.

Averaging hides the failure; the pair adversary exposes it. Table 2 and Figure 1 show the mechanism. SPPO-avg collapses almost entirely onto the decoy: its average win rate is the best of all methods while its minimum falls to 0.060—the hidden-objective failure the construction was designed to reveal, invisible to any average-based metric. Reference

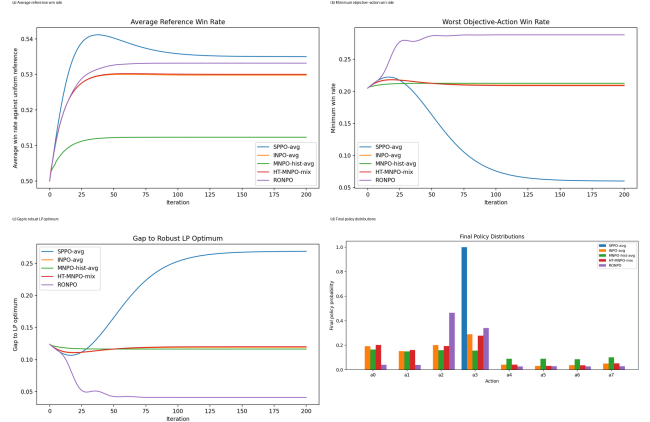


Figure 1: Dynamics over 200 iterations. Averaged-oracle SPPO attains the highest average win rate by collapsing onto the decoy action; RONPO shifts mass toward the robust LP mixture and raises the minimum objective-action win rate.

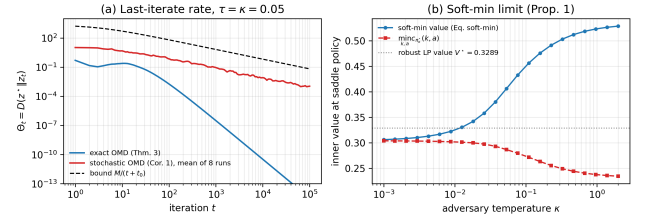


Figure 2: Numerical validation of the theory on the Table 2 instance. (a) Last-iterate Lyapunov value $\Theta_t = D(z^* || z_t)$ under the step schedule of Theorem 2 ($\tau = \kappa = 0.05$). Exact dynamics converge below the explicit bound $M/(t + t_0)$; with single-pair Bernoulli feedback (Cor. 1, mean of 8 runs) the decay is $\Theta(1/t)$ —empirical log-log slope -0.90 over the final decade—running parallel to the bound. (b) The inner soft-min value and the hard minimum $\min_{k,a} c_{\pi^*}(k, a)$ at the saddle policy both converge to the robust LP value $V^* = 0.3289$ as $\kappa \downarrow 0$ (Prop. 1).

anchoring (INPO-avg) and historical mixing (MNPO-hist-avg) soften but do not fix the collapse, since both still optimize the fixed averaged oracle, and HT-MNPO-mix spreads mass across players without restoring a two-player saddle. RONPO raises $V_{\min}$ to 0.288, reducing the LP gap from 0.269 (SPPO-avg) to 0.041, at an average cost of 0.002. The final adversary concentrates on objective 3—the source that rejects the decoy—confirming that it identifies exposing pairs rather than injecting noise. Appendix C sweeps the decoy severity (RONPO’s advantage is largest exactly when the hidden failure is most severe) and the temperature κ , which interpolates between worst-case and averaging behavior as Prop. 1 predicts.

The dynamics track the predicted rates. Figure 2 ties the numerics to the statements. Panel (a) plots Θ_t against the explicit bound $M/(t + t_0)$ of Theorem 2: the exact dynamics stay below the bound at every iterate (converging faster, as

Method	Local RM				IFEval	
	Avg	Worst	WR _B	wWR _B	Strict	Tok.
Base	0.4852	0.4099	–	–	0.4140	610
HT-MNPO (Sky.)	<u>0.5020</u>	<u>0.4735</u>	<u>52.29</u>	<u>47.53</u>	0.3530	1166
HT-MNPO (Ath.)	0.3881	0.3372	43.84	38.02	0.3290	1970
HT-MNPO (Armo)	0.4148	0.3702	45.36	37.87	0.3216	1571
RONPO	0.7025	0.6701	66.05	60.05	<u>0.3993</u>	725

Table 3: Stage-2 summary on 647 held-out prompts. Local RM: per-prompt min–max-normalized reward averaged over the three objectives (Avg), worst objective (Worst), and average/worst win rate vs. the base response (WR_B/wWR_B, %). IFEval: prompt-strict accuracy and mean output tokens. RONPO improves every robustness metric while staying far closer to the base model’s instruction following and length than the single-oracle baselines.

expected for a deterministic map whose fixed point is z^* , while stochastic Bernoulli feedback saturates the $\Theta(1/t)$ rate of Corollary 1. Panel (b) verifies the soft-min characterization: as $\kappa \downarrow 0$ the regularized inner value and the hard pair minimum at the saddle policy converge together to the LP value V^* , confirming both the closed form (9) and the interpretation of κ as the robustness radius. All curves are produced by a self-contained script included with the supplementary code.

LLM Alignment Experiments

Setup. All policies are initialized from Qwen2.5-1.5B-Instruct and trained on 19,856 UltraFeedback prompts with three heterogeneous reward objectives—Skywork, Athene, and ArmoRM (8B-scale reward models; Appendix D)—used through the plug-in oracle (20). Training proceeds in two stages of iterative pool construction and optimization; the response pool $\mathcal{A}(x)$ mixes reference-, current-, and historical-policy generations. Evaluation uses 647 held-out prompts and three axes: (i) the three *local reward models* (training-aligned, so reported as an in-distribution robustness probe: per-prompt normalized reward, worst-objective reward, and win rate against the base response); (ii) *IFEval* (541 prompts, deterministic decoding; Zhou et al. 2023), testing whether preference optimization damages verifiable instruction following; and (iii) a reward-model-independent gpt-5.5 all-pairwise judge with deterministically randomized order, ties counted as 0.5, and prompt-level bootstrap confidence intervals (protocol in Appendix D).

Baselines. We compare against (a) the base policy; (b) three single-oracle HT-MNPO policies, each trained on one reward model, which represent the “pick one judge” strategy; and (c) averaged-oracle self-play baselines SPPO-avg and INPO-avg trained on the prompt-wise average of the min–max-normalized objectives. All trained methods share the same base policy, data, and response pools.

Stage-1 results. On the first-stage checkpoints (full tables in Appendix D), RONPO obtains the highest normalized reward and base-relative win rate under *every* objective, with the best average/worst normalized scores (0.889/0.861 vs. 0.812/0.764 for the strongest single-oracle baseline) and the smallest cross-objective spread (0.020), so the gain is not concentrated in one favorable reward model. The averaged-oracle SPPO-avg and INPO-avg runs are weak in this instantiation; we read this as evidence that a fixed average oracle did not confer objective robustness here, not as a general claim about SPPO or INPO.

Stage-2 results. Table 3 summarizes the final stage. RONPO reaches 0.703/0.670 average/worst normalized reward versus 0.502/0.474 for the strongest single-oracle baseline, with a 66.0% average win rate over the base response. On IFEval the base model remains best, so this axis measures *preservation*: RONPO stays near the base prompt-strict score with moderate output length, while the single-oracle baselines degrade more and generate much longer outputs—so the reward gains are not a length artifact. On the reward-model-independent GPT-5.5 all-pairwise tournament (Appendix D), RONPO ranks first overall, beating every trained baseline clearly (58.7–68.6% head-to-head) while reaching only parity with the base policy (51.8%, CI overlapping 0.5). The combined claim is deliberately scoped: RONPO improves heterogeneous-reward robustness and transfers favorably to an external judge, without uniformly dominating the base model.

Conclusion

RONPO moves the objective index into a single adversary’s action space, keeping heterogeneous preference alignment inside a KL-regularized two-player bilinear saddle: a unique robust Nash policy exists, exact and stochastically fed mirror-descent dynamics converge to it at an $O(1/T)$ last-iterate rate, the inner problem is a KL-DRO worst-case floor over objective–response pairs, and the idealized update is a partition-free log-ratio regression against a sampled adversarial pair. Synthetic games show that averaging can hide a minority-objective collapse that RONPO’s floor prevents and that the dynamics track the predicted rates; at model scale the deployed surrogate improves worst-objective reward and transfers to an external judge.

Limitations. The convergence guarantees are exact for the tabular per-context dynamics and their stochastically fed variant; the prompt-conditioned trainer uses a shared network, a mass-truncated adversary, and the plug-in oracle (20), and its robustness is relative to the finite pool $\mathcal{A}$ and the K oracles. Because the plug-in oracle is Bradley–Terry, model-scale claims concern heterogeneous-*reward* robustness; intransitivity is exercised only in the synthetic games. Model-scale runs currently use a single training seed and a single 1.5B backbone with prompt-level (not seed-level) confidence intervals, and the local reward judges overlap the training objectives; the external judge is the only reward-model-independent signal and shows parity with the base policy on the hardest pairwise subset. Validating the effect on larger backbones, against direct-alignment baselines, across

seeds, and under explicitly opposed objectives (helpfulness vs. conciseness vs. safety) is required before strong empirical claims can be made, and we flag these as open items rather than claiming them.

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A Deferred Proofs

Throughout, $z = (\pi, s)$ and $z' = (\pi', s')$ denote interior points of $Z = \Delta(\mathcal{Y}) \times \Delta(\mathcal{K} \times \mathcal{A})$, and $\delta\pi := \pi - \pi'$, $\delta s := s - s'$.

A.1 Proof of Lemma 1 (Monotonicity Gap)

Because z and z' are interior, every coordinate of π, π', s, s' is strictly positive, so $\log \pi, \log \pi', \log s, \log s'$ are finite, the operator F of (8) is well defined at both points, and every Kullback–Leibler term below is finite. Equip $Z \subseteq \mathbb{R}^{|\mathcal{Y}|} \times \mathbb{R}^{K|\mathcal{A}|}$ with the product inner product

$$\langle (u_1, v_1), (u_2, v_2) \rangle = \langle u_1, u_2 \rangle + \langle v_1, v_2 \rangle. \quad (23)$$

Step 1 (Additive split of the operator). By (8), decompose $F = B + g$ with

$$\begin{aligned} B(\pi, s) &:= (-As, A^\top \pi), \\ g(\pi, s) &:= (\tau(\log(\pi/\mu) + \mathbf{1}), \kappa(\log(s/s_0) + \mathbf{1})). \end{aligned}$$

The map B is linear on Z with block representation

$$B = \begin{pmatrix} 0 & -A \\ A^\top & 0 \end{pmatrix}. \quad (24)$$

Subtracting $F(z')$ from $F(z)$ and using linearity of B ,

$$F(z) - F(z') = B(\delta\pi, \delta s) + (g(z) - g(z')). \quad (25)$$

In $g(z) - g(z')$, the reference vectors $-\tau \log \mu, -\kappa \log s_0$ and the all-ones vectors cancel, leaving

$$g(z) - g(z') = (\tau(\log \pi - \log \pi'), \kappa(\log s - \log s')). \quad (26)$$

Step 2 (The bilinear coupling cancels). By (24) and (23),

$$\langle B(\delta\pi, \delta s), (\delta\pi, \delta s) \rangle = -\langle A \delta s, \delta\pi \rangle + \langle A^\top \delta\pi, \delta s \rangle.$$

Expanding entrywise, with $A_{y,(k,a)} = P_k(y \succ a)$,

$$\langle A \delta s, \delta\pi \rangle = \sum_y \sum_{k,a} A_{y,(k,a)} \delta\pi_y \delta s_{k,a} = \langle A^\top \delta\pi, \delta s \rangle,$$

the same finite double sum in a different order (the adjoint property of the transpose). Carrying opposite signs, the two terms cancel:

$$\langle B(\delta\pi, \delta s), (\delta\pi, \delta s) \rangle = 0. \quad (27)$$

Nothing about A was used beyond its being a fixed matrix, so the coupling vanishes for every collection of oracles $P_1, \dots, P_K$ and every pool $\mathcal{A}$; structurally, (27) says B is skew-symmetric, $B^\top = -B$.

Step 3 (The regularizer yields a symmetrized KL). Pairing (26) against $(\delta\pi, \delta s)$,

$$\begin{aligned} \langle g(z) - g(z'), (\delta\pi, \delta s) \rangle &= \tau \langle \log \pi - \log \pi', \pi - \pi' \rangle \\ &\quad + \kappa \langle \log s - \log s', s - s' \rangle. \end{aligned} \quad (28)$$

Splitting the policy-block sum into its $\pi(y)$ - and $\pi'(y)$ -weighted parts,

$$\begin{aligned} \langle \log \pi - \log \pi', \pi - \pi' \rangle &= \sum_y \pi(y) (\log \pi(y) - \log \pi'(y)) \\ &\quad + \sum_y \pi'(y) (\log \pi'(y) - \log \pi(y)) \\ &= \text{KL}(\pi \| \pi') + \text{KL}(\pi' \| \pi), \end{aligned}$$

and identically for the adversary block. Substituting into (28),

$$\begin{aligned} \langle g(z) - g(z'), (\delta\pi, \delta s) \rangle &= \tau \{ \text{KL}(\pi \| \pi') + \text{KL}(\pi' \| \pi) \} \\ &\quad + \kappa \{ \text{KL}(s \| s') + \text{KL}(s' \| s) \}. \end{aligned} \quad (29)$$

Step 4 (Assembling). Pairing (25) against $z - z'$ and using (27) and (29) gives (10).

Step 5 (Strong-monotonicity bound). Relative entropy is nonnegative, so replacing τ and κ by $m = \min\{\tau, \kappa\}$ in front of nonnegative terms can only decrease the value:

$$\langle F(z) - F(z'), z - z' \rangle \geq m \{ D(z \| z') + D(z' \| z) \},$$

with $D(z \| z') = \text{KL}(\pi \| \pi') + \text{KL}(s \| s')$. $\square$

A.2 Proof of Theorem 1 (Existence and Uniqueness)

For fixed s , $\pi \mapsto V(\pi, s)$ is strictly concave (linear minus $\tau \text{KL}(\pi \| \mu)$), with unique best response

$$B_\pi(s)(y) = \frac{\mu(y) \exp((As)_y / \tau)}{\sum_u \mu(u) \exp((As)_u / \tau)}. \quad (30)$$

For fixed π , $s \mapsto V(\pi, s)$ is strictly convex, with unique best response

$$B_s(\pi)(k, a) = \frac{s_0(k, a) \exp(-(A^\top \pi)_{k,a} / \kappa)}{\sum_{j,b} s_0(j, b) \exp(-(A^\top \pi)_{j,b} / \kappa)}. \quad (31)$$

The map $(\pi, s) \mapsto (B_\pi(s), B_s(\pi))$ is continuous on the compact convex product of simplices, so Brouwer’s theorem gives a fixed point (π^*, s^*) with $\pi^* = B_\pi(s^*)$ and $s^* = B_s(\pi^*)$. Since π^* is the exact policy best response to s^* and s^* the exact adversary best response to π^* ,

$$V(\pi, s^*) \leq V(\pi^*, s^*) \leq V(\pi^*, s) \quad \forall \pi, s, \quad (32)$$

which is the definition of a saddle point. Every saddle point must satisfy the same best-response equations, and since

μ, s_0 are strictly positive and the exponentials in (30)–(31) are strictly positive, every saddle point has full support.

For uniqueness, we first derive the saddle variational inequality. Fix any feasible π and set $\pi_\epsilon := \pi^* + \epsilon(\pi - \pi^*)$ for $\epsilon \in [0, 1]$; feasibility follows from convexity of the simplex. Since π^* maximizes $V(\cdot, s^*)$,

$$\frac{V(\pi_\epsilon, s^*) - V(\pi^*, s^*)}{\epsilon} \leq 0 \implies \langle \nabla_\pi V(z^*), \pi - \pi^* \rangle \leq 0$$

by letting $\epsilon \downarrow 0$ and using differentiability at the interior point z^* . Likewise, minimality of s^* for $V(\pi^*, \cdot)$ gives $\langle \nabla_s V(z^*), s - s^* \rangle \geq 0$ for all feasible s . Negating the first inequality, adding the second, and using $F = (-\nabla_\pi V, \nabla_s V)$ yields

$$\langle F(z^*), z - z^* \rangle \geq 0 \quad \forall z \in Z. \quad (33)$$

Now let z_1, z_2 be two saddle points. Applying (33) at z_1 with $z = z_2$ and at z_2 with $z = z_1$, then adding,

$$\langle F(z_1) - F(z_2), z_1 - z_2 \rangle \leq 0.$$

Lemma 1 lower-bounds the same quantity by $m\{D(z_1\|z_2) + D(z_2\|z_1)\} \geq 0$. Both hold only if the divergences vanish, hence $z_1 = z_2$. $\square$

A.3 Proof of Proposition 1 and Remark 1

For fixed π , $\min_s V(\pi, s) = -\tau \text{KL}(\pi\|\mu) + \min_s \{ \langle c_\pi, s \rangle + \kappa \text{KL}(s\|s_0) \}$. The inner minimization is the Gibbs variational principle: for any cost vector c and prior s_0 ,

$$\min_{s \in \Delta} \{ \langle c, s \rangle + \kappa \text{KL}(s\|s_0) \} = -\kappa \log \mathbb{E}_{s_0}[e^{-c/\kappa}],$$

attained uniquely at $s^* \propto s_0 \exp(-c/\kappa)$, obtained by setting the gradient of the strictly convex objective to a constant on the simplex. Substituting $c = c_\pi$ gives (9). The limit $\kappa \downarrow 0$ is the standard log-sum-exp $\rightarrow \min$ convergence: $-\kappa \log \sum_i s_0(i) e^{-c_i/\kappa} \rightarrow \min_i c_i$ for s_0 with full support.

For Remark 1: for any $k \in \mathcal{K}$ and $q \in \Delta(\mathcal{A})$, $\mathbb{E}_{a \sim q}[c_\pi(k, a)] \geq \min_{a \in \mathcal{A}} c_\pi(k, a) \geq \min_{(k', a) \in \mathcal{K} \times \mathcal{A}} c_\pi(k', a)$, since an average dominates a minimum and a minimum over a larger set is no larger. Taking the minimum over k preserves the inequality. $\square$

A.4 Proof of Lemma 2 (Operator Bound)

We bound the policy block $F_\pi(z_t) = -As_t + \tau(\psi_t + \mathbf{1})$ with $\psi_t := \log(\pi_t/\mu)$; the adversary block is symmetric. Since $P_k \in [0, 1]$ and s_t is a distribution, $(As_t)_y = \mathbb{E}_{(k, a) \sim s_t}[P_k(y \succ a)] \in [0, 1]$, so $\|As_t\|_\infty \leq 1$.

Normalization. Because π_t and μ are distributions, $\mathbb{E}_\mu[e^{\psi_t}] = \sum_y \mu(y) \pi_t(y) / \mu(y) = 1$, so ψ_t is neither everywhere positive nor everywhere negative: $\min_y \psi_t(y) \leq 0 \leq \max_y \psi_t(y)$.

Spread. Taking logs of (14), $\psi_{t+1}(y) = \eta_t r_t(y) + (1 - \eta_t \tau) \psi_t(y) - \log Z_t$ with normalizer Z_t ; the constant cancels in differences, so for all y, y' , using $1 - \eta_t \tau \geq 0$,

$$\begin{aligned} \psi_{t+1}(y) - \psi_{t+1}(y') &= \eta_t (r_t(y) - r_t(y')) \\ &\quad + (1 - \eta_t \tau) (\psi_t(y) - \psi_t(y')). \end{aligned}$$

Let $\sigma_t := \max_y \psi_t(y) - \min_y \psi_t(y)$. As $r_t(y) \in [0, 1]$, its spread is at most 1, so $\sigma_{t+1} \leq \eta_t + (1 - \eta_t \tau) \sigma_t$. From $\pi_0 = \mu$

(so $\sigma_0 = 0$), induction yields $\sigma_t \leq 1/\tau$: if $\sigma_t \leq 1/\tau$ then $\sigma_{t+1} \leq \eta_t + (1 - \eta_t \tau)/\tau = 1/\tau$.

Combine. From $\min_y \psi_t \leq 0 \leq \max_y \psi_t$ and $\sigma_t \leq 1/\tau$ we get $\|\psi_t\|_\infty \leq 1/\tau$, hence

$$\begin{aligned} \|F_\pi(z_t)\|_\infty &\leq \|As_t\|_\infty + \tau \|\psi_t + \mathbf{1}\|_\infty \\ &\leq 1 + \tau(\frac{1}{\tau} + 1) = 2 + \tau. \end{aligned}$$

The identical argument for $\log(s_t/s_0)$ gives $\|F_s(z_t)\|_\infty \leq 2 + \kappa$; taking the maximum gives $G = 2 + \max\{\tau, \kappa\}$. Note that only $r_t \in [0, 1]^{|Y|}$ and $c_t \in [0, 1]^{K|A|}$ coordinatewise were used, a fact we reuse in Corollary 1. $\square$

A.5 Proof of Theorem 2 (Last-Iterate Rate)

Apply the descent form of the one-step entropic-OMD inequality (41) separately to the two blocks: policy block with $u_t = \pi_t$, comparator π^* , gradient $F_\pi(z_t)$; adversary block with $u_t = s_t$, comparator s^* , gradient $F_s(z_t)$. Adding,

$$\begin{aligned} \Theta_{t+1} &\leq \Theta_t - \eta_t \langle F(z_t), z_t - z^* \rangle \\ &\quad + \frac{\eta_t^2}{2} (\|F_\pi(z_t)\|_\infty^2 + \|F_s(z_t)\|_\infty^2), \end{aligned} \quad (34)$$

where the two middle terms combined via the product inner product. Lemma 2 bounds the last term by $\eta_t^2 G^2$. Decompose the inner product as $\langle F(z_t) - F(z^*), z_t - z^* \rangle + \langle F(z^*), z_t - z^* \rangle$: the variational inequality (33) makes the second term nonnegative, and Lemma 1 bounds the first below by $m\{D(z_t\|z^*) + D(z^*\|z_t)\} \geq m\Theta_t$. This gives (16).

It remains to solve the recurrence. Set $a_t := \Theta_t$ and $C := 4G^2/m^2$. With $\eta_t = 2/(m(t + t_0))$, (16) reads

$$a_{t+1} \leq \left(1 - \frac{2}{t + t_0}\right) a_t + \frac{C}{(t + t_0)^2}.$$

Let $M := \max\{t_0 a_0, C\}$. We show $a_t \leq M/(t + t_0)$ by induction: the base case is $a_0 \leq M/t_0$; for the step, write $n = t + t_0$, so

$$\begin{aligned} a_{t+1} &\leq \left(1 - \frac{2}{n}\right) \frac{M}{n} + \frac{C}{n^2} = \frac{M(n-2) + C}{n^2} \\ &\leq \frac{M(n-1)}{n^2} \leq \frac{M}{n+1}, \end{aligned}$$

using $C \leq M$ and $(n-1)(n+1) \leq n^2$. $\square$

A.6 Proof of Corollary 1 (Stochastic Feedback)

Let $\mathcal{F}_t$ be the σ -field generated by the history through z_t . The realized gradients are $\hat{F}_\pi(z_t) = -\hat{r}_t + \tau(\psi_t + \mathbf{1})$ and $\hat{F}_s(z_t) = \hat{c}_t + \kappa(\log(s_t/s_0) + \mathbf{1})$.

Pathwise bound. By construction $\hat{r}_t(y) = \xi_t(y) \in \{0, 1\} \subset [0, 1]$ and $\hat{c}_t(k, a) = \zeta_t(k, a) \in \{0, 1\} \subset [0, 1]$ coordinatewise. The spread induction in the proof of Lemma 2 used only that the reward vector lies in $[0, 1]$ coordinatewise, so it applies verbatim to the stochastic iterates and yields, pathwise, $\|\hat{F}_\pi(z_t)\|_\infty \leq G$ and $\|\hat{F}_s(z_t)\|_\infty \leq G$.

Unbiasedness. Conditional on $\mathcal{F}_t$, $\mathbb{E}[\xi_t(y)] = \mathbb{E}_{(k, a) \sim s_t}[P_k(y \succ a)] = r_t(y)$ for every y , and $\mathbb{E}[\zeta_t(k, a)] = \mathbb{E}_{y \sim \pi_t}[P_k(y \succ a)] = c_t(k, a)$ for every (k, a) ; hence $\mathbb{E}[\hat{F}(z_t) \mid \mathcal{F}_t] = F(z_t)$, since the regularizer terms are $\mathcal{F}_t$ -measurable.

Recurrence in expectation. The one-step inequality (41) is a deterministic consequence of the update and holds pathwise with the realized gradients:

$$\Theta_{t+1} \leq \Theta_t - \eta_t \langle \widehat{F}(z_t), z_t - z^* \rangle + \eta_t^2 G^2.$$

Taking $\mathbb{E}[\cdot | \mathcal{F}_t]$ and using that z_t, z^* are $\mathcal{F}_t$ -measurable,

$$\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] \leq \Theta_t - \eta_t \langle F(z_t), z_t - z^* \rangle + \eta_t^2 G^2,$$

and the deterministic decomposition from the proof of Theorem 2 (variational inequality plus Lemma 1) bounds the middle term to give $\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] \leq (1 - m\eta_t)\Theta_t + \eta_t^2 G^2$. The tower property yields the same scalar recurrence for $\mathbb{E}[\Theta_t]$, which the induction of Theorem 2 solves to $\mathbb{E}[\Theta_T] \leq M/(T + t_0)$. Each round uses one Bernoulli query per coordinate: $|\mathcal{Y}|$ queries against the sampled pair (k_t, a_t) and $K|\mathcal{A}|$ queries at the sampled response y_t . $\square$

A.7 Proof of Proposition 2 (Unbiased Regression)

Fix an ordered pair (y, y') and define $d_t(y, y') := r_t(y) - r_t(y')$. With $N = z_y - z_{y'}$, conditioning on (y, y') leaves randomness only in the sampled pair $(k, a) \sim s_t$ and the two Bernoulli draws, so

$$\begin{aligned} \mathbb{E}[N | y, y'] &= \mathbb{E}_{(k,a) \sim s_t} [P_k(y \succ a) - P_k(y' \succ a)] \\ &= d_t(y, y'). \end{aligned}$$

For any candidate π , adding and subtracting $\eta_t d_t(y, y')$ inside the square and using that the cross term vanishes in conditional expectation,

$$\begin{aligned} \mathbb{E}[(h_t(\pi; y, y') - \eta_t N)^2 | y, y'] \\ = (h_t(\pi; y, y') - \eta_t d_t(y, y'))^2 + \eta_t^2 \text{Var}(N | y, y'). \end{aligned}$$

The variance term does not depend on π , so the conditional loss is minimized exactly when $h_t(\pi; y, y') = \eta_t d_t(y, y')$, which the exact update π_{t+1} satisfies for every pair by (18); hence π_{t+1} minimizes the population loss. If π_t has full support, every ordered pair has positive sampling probability, so any other minimizer π must satisfy $h_t(\pi; y, y') = h_t(\pi_{t+1}; y, y')$ for all pairs; the μ and π_t terms cancel, leaving $\Delta_\pi(y, y') = \Delta_{\pi_{t+1}}(y, y')$ for all y, y' . Fixing y_0 and taking $y' = y_0$ shows $\log \pi(y) - \log \pi_{t+1}(y)$ is constant in y , so $\pi = C\pi_{t+1}$, and normalization forces $C = 1$. $\square$

B Standard Online-Mirror-Descent Facts

The building blocks below are classical (Cesa-Bianchi and Lugosi 2006; Bubeck 2015; Nemirovski and Yudin 1983); we include them so the paper is self-contained. We take Pinsker's and Hölder's inequalities as given and derive the regret-to-gap reduction, the one-step inequality (41), and the regret bound used in the section on no-regret learning.

B.1 Regret-to-Gap Reduction

At round t , let the maximizing player use π_t and the minimizing player use π'_t in the game (2). Once π'_t is fixed, the maximizing player faces the convex online loss $\ell_t^\pi(\pi) := -J(\pi, \pi'_t)$; once π_t is fixed, the minimizing player faces

$\ell_t^{\pi'}(\pi') := J(\pi_t, \pi')$. Convexity holds because $J(\cdot, \pi'_t)$ is concave and $J(\pi_t, \cdot)$ is convex. The external regrets are

$$\begin{aligned} R_T^\pi &= \max_{\pi \in \Delta(\mathcal{Y})} \sum_{t=1}^T J(\pi, \pi'_t) - \sum_{t=1}^T J(\pi_t, \pi'_t), \\ R_T^{\pi'} &= \sum_{t=1}^T J(\pi_t, \pi'_t) - \min_{\pi' \in \Delta(\mathcal{Y})} \sum_{t=1}^T J(\pi_t, \pi'). \end{aligned}$$

Let $\bar{\pi}_T = \frac{1}{T} \sum_{t=1}^T \pi_t$ and $\bar{\pi}'_T = \frac{1}{T} \sum_{t=1}^T \pi'_t$, and define the gap $\text{Gap}(\bar{\pi}_T, \bar{\pi}'_T) = \max_{\pi} J(\pi, \bar{\pi}'_T) - \min_{\pi'} J(\bar{\pi}_T, \pi')$. For fixed π , convexity of $J(\pi, \cdot)$ and Jensen give $J(\pi, \bar{\pi}'_T) \leq \frac{1}{T} \sum_{t=1}^T J(\pi, \pi'_t)$; for fixed π' , concavity gives $J(\bar{\pi}_T, \pi') \geq \frac{1}{T} \sum_{t=1}^T J(\pi_t, \pi')$. Taking the maximum in the first, the minimum in the second, and adding and subtracting $\sum_{t=1}^T J(\pi_t, \pi'_t)$,

$$\text{Gap}(\bar{\pi}_T, \bar{\pi}'_T) \leq \frac{R_T^\pi + R_T^{\pi'}}{T}. \quad (35)$$

Thus if both players are no-regret, the gap of average play vanishes; this is the precise game-theoretic reason for using no-regret dynamics rather than arbitrary stabilized best responses. Sublinear regret certifies average iterates only; RONPO's last-iterate results use the additional strong monotonicity of Lemma 1.

B.2 One-Step Inequality and Regret Bound

Let $\ell_1, \dots, \ell_T$ be convex losses on $\Delta(\mathcal{Y})$, let u_{t+1} be the entropic-OMD iterate (3) from u_t with step $\eta_t > 0$, let $g_t \in \partial \ell_t(u_t)$, and fix a comparator $\tilde{u} \in \Delta(\mathcal{Y})$; write $n = |\mathcal{Y}|$.

Step 0 (Linearization). By the subgradient inequality, $\ell_t(u_t) - \ell_t(\tilde{u}) \leq \langle g_t, u_t - \tilde{u} \rangle$ (equality for linear losses, as in the RONPO blocks); summing, it suffices to bound each $\langle g_t, u_t - \tilde{u} \rangle$.

Step 1 (Closed form and log-linear relation). The subproblem (3) has the multiplicative-weights closed form (4) (App. B.3); taking logarithms,

$$\eta_t g_t = \log(u_t / u_{t+1}) - (\log Z_t) \mathbf{1} \quad (36)$$

componentwise, with all logarithms finite since interiority is preserved.

Step 2 (Three-point identity). Pair (36) with $u_{t+1} - \tilde{u}$; the constant drops because $\langle \mathbf{1}, u_{t+1} - \tilde{u} \rangle = 0$. Splitting the resulting sum into its u_{t+1} - and $\tilde{u}$ -weighted parts and using $\log(u_t / u_{t+1}) = \log(\tilde{u} / u_{t+1}) - \log(\tilde{u} / u_t)$,

$$\begin{aligned} \eta_t \langle g_t, u_{t+1} - \tilde{u} \rangle &= \text{KL}(\tilde{u} \| u_t) - \text{KL}(\tilde{u} \| u_{t+1}) \\ &\quad - \text{KL}(u_{t+1} \| u_t). \end{aligned} \quad (37)$$

Step 3 (Splitting the per-step term). Inserting u_{t+1} as an intermediate point and substituting (37),

$$\begin{aligned} \eta_t \langle g_t, u_t - \tilde{u} \rangle &= \eta_t \langle g_t, u_t - u_{t+1} \rangle + \text{KL}(\tilde{u} \| u_t) \\ &\quad - \text{KL}(\tilde{u} \| u_{t+1}) - \text{KL}(u_{t+1} \| u_t). \end{aligned} \quad (38)$$

Step 4 (Hölder, Pinsker, Young). Hölder gives $\eta_t \langle g_t, u_t - u_{t+1} \rangle \leq \eta_t \|g_t\|_\infty \|u_t - u_{t+1}\|_1$ and Pinsker gives $-\text{KL}(u_{t+1} \| u_t) \leq -\frac{1}{2} \|u_t - u_{t+1}\|_1^2$; with $a = \eta_t \|g_t\|_\infty$ and $b = \|u_t - u_{t+1}\|_1$, the elementary bound $ab - \frac{1}{2} b^2 \leq \frac{1}{2} a^2$ yields

$$\eta_t \langle g_t, u_t - u_{t+1} \rangle - \text{KL}(u_{t+1} \| u_t) \leq \frac{1}{2} \eta_t^2 \|g_t\|_\infty^2. \quad (39)$$

Step 5 (One-step inequality). Substituting (39) into (38) and dividing by η_t ,

$$\langle g_t, u_t - \tilde{u} \rangle \leq \frac{\text{KL}(\tilde{u} \| u_t) - \text{KL}(\tilde{u} \| u_{t+1})}{\eta_t} + \frac{\eta_t}{2} \|g_t\|_\infty^2; \quad (40)$$

multiplying back by η_t and rearranging gives the descent form

$$\text{KL}(\tilde{u} \| u_{t+1}) \leq \text{KL}(\tilde{u} \| u_t) - \eta_t \langle g_t, u_t - \tilde{u} \rangle + \frac{1}{2} \eta_t^2 \|g_t\|_\infty^2, \quad (41)$$

the per-block step invoked in Theorem 2 and Corollary 1.

Step 6 (Regret at constant step). With $\eta_t \equiv \eta$, summing (40) telescopes the KL terms; using $\text{KL}(\tilde{u} \| u_1) \leq \log n$ for uniform u_1 and $\|g_t\|_\infty \leq G$,

$$R_T \leq \frac{\log n}{\eta} + \frac{\eta}{2} G^2 T \leq G \sqrt{2T \log n}$$

at the optimal $\eta = \sqrt{2 \log n / (G^2 T)}$, so the dynamics are no-regret.

B.3 Multiplicative-Weights Mirror Step

The subproblem (3) has a strictly convex objective with a full-support reference u_t , so its minimizer is interior and unique (the entropy gradient $\log v(y)$ diverges as $v(y) \downarrow 0$, so no coordinate is set to zero). Introducing a multiplier λ for $\sum_y v(y) = 1$ gives the stationarity condition $\eta_t g_t(y) + \log(v(y)/u_t(y)) + 1 + \lambda = 0$, hence $v(y) \propto u_t(y) e^{-\eta_t g_t(y)}$; normalizing recovers (4). Since u_1 is interior and each step preserves full support, every iterate is interior.

C Synthetic-Game Details and Additional Figures

Game construction. Each objective $k \in \{1, 2, 3\}$ is an 8×8 skew-symmetric matrix built in three layers: (i) a weak random tournament background assigning each ordered pair win probability 0.56 or 0.44 uniformly at random; (ii) a specialist action a_k for objective k that beats every other action with probability 0.82 under P_k ; (iii) a decoy action that beats every other action with probability 0.78 under objectives 1, 2 and 0.06 under objective 3. The reference μ and adversary prior s_0 are uniform. The robust LP benchmark $V^* = \max_\pi \min_{k,a} c_\pi(k, a)$ is computed by linear programming; the reported instance has $V^* = 0.3289$.

Decoy severity sweep. Figure 3 varies the hidden-objective decoy win probability. When the decoy is most dangerous, averaged-oracle methods keep assigning it high mass because it remains attractive under the first two objectives, and RONPO’s advantage is largest; as the hidden objective becomes milder, robust and average objectives align and the gap narrows.

Temperature sweep. Figure 4 traces the average/worst-case trade-off as κ varies: small κ sharpens the worst-case adversary; large κ keeps the adversary near its prior and recovers averaging behavior, the empirical counterpart of the interpolation in Prop. 1.

D LLM Experiment Details

D.1 Stage-1 Evaluation

Compared models and training signals. All methods are initialized from Qwen/Qwen2.5-1.5B-Instruct. Table 4 lists the compared stage-1 policies and the preference signal used to train each. HT-MNPO and RONPO share the same stage-1 base-policy response pool; SPPO-avg and INPO-avg use the prompt-wise average of the min-max-normalized objectives.

Training configuration. Table 5 summarizes the common stage-1 configuration for HT-MNPO and RONPO; Table 6 gives method-specific settings. The SPPO-avg and INPO-avg artifacts record the checkpoint and oracle construction but not the complete optimizer configuration; we do not infer unrecorded hyperparameters.

Evaluation configuration and metrics. For evaluation, every model generates one response per prompt under identical vLLM decoding (seed 42, temperature 0.7, top- p 0.9, at most 2048 new tokens), and all outputs are rescored by the same three reward models in one pass (Table 8, left). Let $r_{i,o,m}$ be the score on prompt i from objective o for model m . Because reward scales differ, we report per-prompt min-max-normalized scores

$$\hat{r}_{i,o,m} = \frac{r_{i,o,m} - \min_{m'} r_{i,o,m'}}{\max_{m'} r_{i,o,m'} - \min_{m'} r_{i,o,m'}}, \quad (42)$$

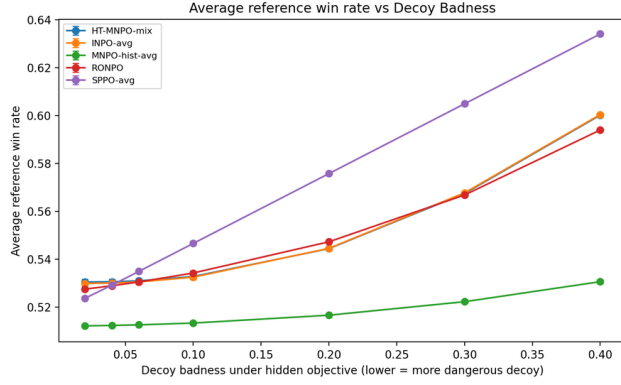
with $\hat{r}_{i,o,m} = 1/2$ when the denominator is zero, averaged over prompts; the extrema range over the exact compared model set, so normalized values are comparable only within one table. Win rates against the base use $w = 1$ if $r_{i,o,m} > r_{i,o,m_0}$, $w = 0$ if smaller, and $w = 1/2$ on exact ties. Average metrics take the mean over objectives; worst-objective metrics take the minimum; the normalized standard deviation is computed across the three objective-level normalized scores.

D.2 Stage-2 Evaluation

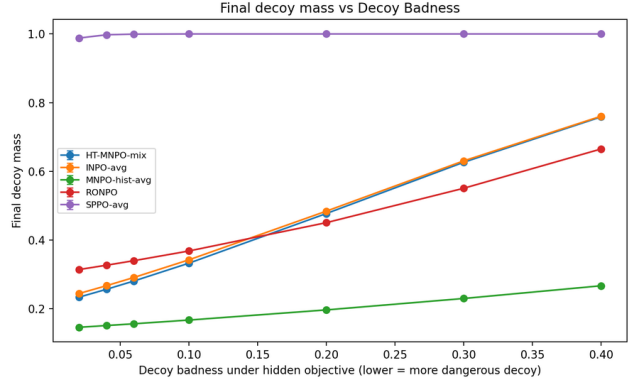
Compared artifacts and setup. Stage 2 resumes training from stage-1 checkpoints; Table 9 lists the compared policies (an earlier diagnostic RONPO stage-2 run is excluded because it exhibited output-length pathology; the resumed run reported here does not). Table 10 gives the evaluation configuration. The local reward-model and GPT-5.5 evaluations share the same generated responses, so they differ only in the evaluator; IFEval measures rule-based instruction following on a separate benchmark with deterministic decoding.

External-judge protocol. The judge receives the prompt and two candidate responses labeled A and B, in a deterministically randomized order seeded by the prompt hash, and returns a single preference (A, B, or tie) under a fixed rubric covering helpfulness, correctness, and instruction following. Each unordered pair is judged once per prompt; a tie counts as 0.5 for both sides. Mean pairwise win rates and 95% intervals use 2000 prompt-level paired bootstrap resamples. Because the judge is a proprietary model, exact reproduction depends on its availability; we recommend re-running

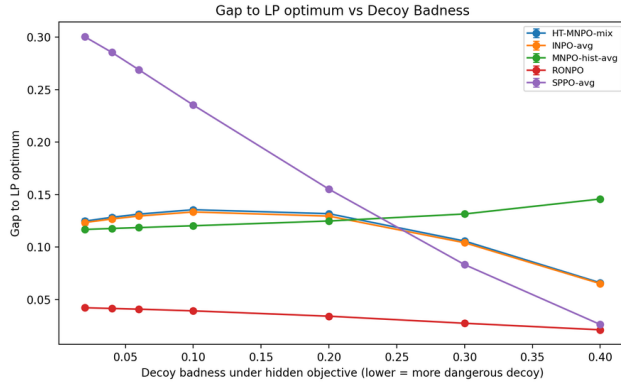
(a) Average reference win rate



(b) Final decoy mass



(c) Gap to robust LP optimum



(d) Minimum win rate

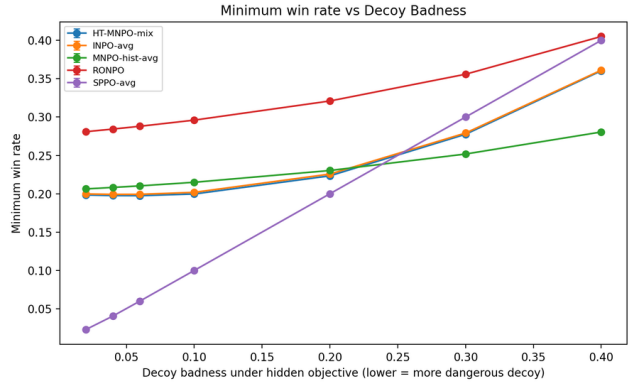


Figure 3: Sweep over the hidden-objective decoy win probability $P_3(a_{\text{decoy}} \succ a)$. Lower values make the decoy more dangerous. RONPO is most useful in the regime where the averaged oracle masks severe minority-objective failure.

the protocol with an open-weight judge for an independent check.

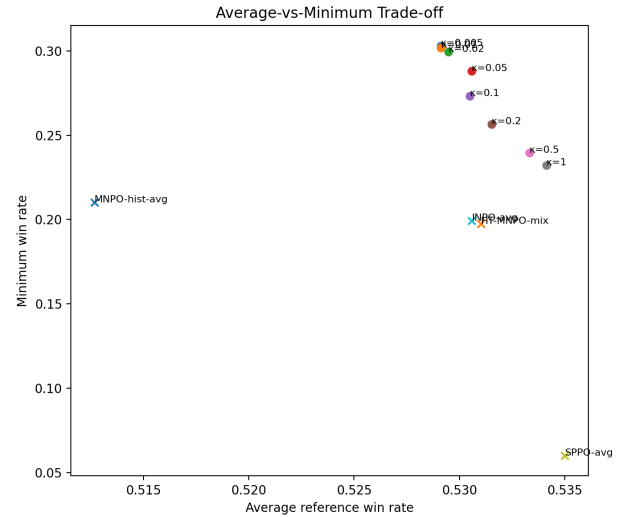


Figure 4: RONPO trade-off as the adversary temperature κ varies. Smaller κ gives a sharper worst-case adversary; larger κ moves toward fixed scalarization.

Method	Evaluated checkpoint	Training signal
Base	Qwen/Qwen2.5-1.5B-Instruct	No stage-1 preference optimization
HT-MNPO (Sky.)	Final stage-1 checkpoint	Homogeneous Skywork oracle
HT-MNPO (Ath.)	Stage-1 checkpoint 300	Homogeneous Athene oracle
HT-MNPO (Armo)	Final stage-1 checkpoint	Homogeneous ArmoRM oracle
RONPO	Stage-1 checkpoint 1100	Adaptive adversary over Skywork, Athene, and ArmoRM
SPPO-avg	Final stage-1 checkpoint	Prompt-wise average of normalized objectives
INPO-avg	Final stage-1 checkpoint	Prompt-wise average of normalized objectives

Table 4: Compared stage-1 policies and their training signals.

Setting	Value
Base policy	Qwen/Qwen2.5-1.5B-Instruct
Training split	UltraFeedback subset; 19,856 prompts
Held-out split	UltraFeedback subset; 647 prompts
Online generation	vLLM with seeds 13, 21, 42, 79, 100, temperature 0.8, top- p 0.95, at most 4096 new tokens
Objectives	Skywork, Athene, and ArmoRM
Shared stage-1 data	HT-MNPO and RONPO use the same base-policy response pool
Optimization	bf16 AdamW, learning rate 5×10^{-7} , cosine schedule, warmup ratio 0.1, weight decay 0
Training horizon	One epoch; seed 42
Batching	Per-device batch size 2; gradient accumulation 8
Sequence limits	Maximum sequence length 2048; maximum prompt length 1800
Memory and launch	Gradient checkpointing (<code>use_reentrant=false</code>); Accelerate with DeepSpeed ZeRO-3
Checkpointing	Evaluation and saving every 100 steps; save-total limit 5

Table 5: Common stage-1 training configuration for HT-MNPO and RONPO.

Setting	HT-MNPO stage 1	RONPO stage 1
Objective handling	One homogeneous oracle per run	Skywork, Athene, and ArmoRM jointly
Pair or target construction	Reward-gap target; no target normalization	Per-objective min-max normalization; adversarial pair selection; best-vs-adversary policy pair
Method scales	Ratio 0.3333; $\eta = 0.0075$; $\beta = (10, 1, 10)$ for Skywork, Athene, ArmoRM	Policy $\alpha = 1.0$; $\tau = 0.05$; preference scale $\lambda = 8.0$; $\beta = 10$
Adversary settings	Not applicable	25 adversary steps; adversary $\alpha = 1.0$; $\kappa = 0.05$; one pair per prompt

Table 6: Method-specific stage-1 training configuration.

Method	Skywork		Athene		ArmoRM		Normalized aggregate			WR aggregate	
	Norm.	WR	Norm.	WR	Norm.	WR	Avg.	Worst	Std.	Avg.	Worst
Base	0.7285	–	0.7501	–	0.8645	–	0.7811	0.7285	0.0597	–	–
HT-MNPO (Sky.)	0.7637	55.95	0.7889	58.19	0.8833	58.73	0.8119	0.7637	0.0515	57.62	55.95
HT-MNPO (Ath.)	0.7640	56.03	0.7557	49.69	0.8609	49.61	0.7935	0.7557	0.0477	51.78	49.61
HT-MNPO (Armo)	0.7552	56.34	0.7531	55.72	0.8494	50.70	0.7859	0.7531	0.0449	54.25	50.70
RONPO	0.8609	67.47	0.8950	68.55	0.9095	59.81	0.8885	0.8609	0.0204	65.28	59.81
SPPO-avg	0.3361	13.60	0.0838	2.78	0.1948	1.08	0.2049	0.0838	0.1032	5.82	1.08
INPO-avg	0.0457	3.86	0.1288	6.57	0.0274	0.77	0.0673	0.0274	0.0441	3.74	0.77

Table 7: Stage-1 local reward-model evaluation on 647 held-out prompts. *Norm.* is the per-prompt normalized reward and *WR* the win rate against the base response (%). Lower normalized standard deviation indicates more balanced objective-level performance. RONPO achieves the best average and worst-objective results under all three reward models.

Objective	Reward model	Opponent	Sky.	Ath.	Armo	Avg.	Worst
Skywork	Skywork/Skywork-Reward-V2-Llama-3.1-8B	HT-MNPO (Sky.)	62.29	63.37	57.50	61.05	57.50
Athene	Nexusflow/Athene-RM-8B	HT-MNPO (Ath.)	63.21	68.32	63.83	65.12	63.21
ArmoRM	RLHFlow/ArmoRM-Llama3-8B-v0.1	HT-MNPO (Armo)	63.68	67.31	62.21	64.40	62.21
		SPPO-avg	92.97	98.84	98.61	96.81	92.97
		INPO-avg	97.53	98.38	99.85	98.58	97.53

Table 8: Stage-1 scoring models (left) and RONPO pairwise win rate (%) against each trained baseline under every local reward objective (right). The evaluation is limited to judges overlapping the training objectives; it does not establish human-preference superiority or cross-seed significance.

Method	Artifact or checkpoint	Training signal
Base	Qwen/Qwen2.5-1.5B-Instruct	No stage-2 preference optimization
HT-MNPO (Sky.)	Skywork stage-2 output	Homogeneous Skywork oracle
HT-MNPO (Ath.)	Athene stage-2 output	Homogeneous Athene oracle
HT-MNPO (Armo)	ArmoRM stage-2 output	Homogeneous ArmoRM oracle
RONPO (ckpt 1400)	Resumed stage-2 checkpoint 1400	Robust objective-response adversary
RONPO (ckpt 2457)	Resumed stage-2 final checkpoint 2457	Robust objective-response adversary

Table 9: Compared stage-2 artifacts. The averaging baselines appear only at stage 1; restoring tuned stage-2 versions is a stated open item.

Setting	Value
Held-out preference split	UltraFeedback subset (disjoint from stage-1 test); 647 prompts
Generation for local RM and GPT judge	vLLM, seed 42, temperature 0.7, top- p 0.9, at most 4096 new tokens, bf16
Local RM objectives	Skywork, Athene, and ArmoRM
Local RM normalization	Per-prompt min-max normalization over the exact six-model stage-2 comparison set
Local RM confidence intervals	2000 prompt-level paired bootstrap resamples
GPT judge	gpt-5.5-2026-04-23 through the batch API
GPT judge comparisons	All 15 unordered model pairs per prompt; response order deterministically randomized
GPT judge coverage	9705/9705 parsed judgments after retrying output-limit failures
GPT judge confidence intervals	2000 prompt-level paired bootstrap resamples
IFEval	EvalScope 1.8.1 on opencompass/ifeval, 541 valid prompts, deterministic decoding

Table 10: Stage-2 evaluation configuration.

Method	Mean pairwise WR	95% CI
RONPO (ckpt 2457)	0.5927	[0.5709, 0.6136]
Base	0.5581	[0.5329, 0.5841]
RONPO (ckpt 1400)	0.5532	[0.5306, 0.5750]
HT-MNPO (Sky.)	0.5054	[0.4841, 0.5272]
HT-MNPO (Armo)	0.4263	[0.4025, 0.4493]
HT-MNPO (Ath.)	0.3643	[0.3425, 0.3855]

Table 11: Reward-model-independent GPT-5.5 all-pairwise judge results on 647 held-out prompts. Each method is compared against the other five; higher mean pairwise win rate is better.

Opponent	Sky.	Ath.	Armo	Avg.	Worst
Base	68.86	69.24	60.05	66.05	60.05
HT-MNPO (Sky.)	63.76	71.79	68.32	67.96	63.76
HT-MNPO (Ath.)	64.76	79.91	73.96	72.88	64.76
HT-MNPO (Armo)	64.30	77.51	73.88	71.90	64.30

Table 12: Local reward-model pairwise win rates (%) of RONPO stage-2 checkpoint 2457 against each comparison model.

Pair	n	Left WR	Tie
Base vs RONPO (2457)	647	0.4822	0.1360
HT-MNPO (Sky.) vs RONPO (2457)	647	0.4134	0.1808
HT-MNPO (Ath.) vs RONPO (2457)	647	0.3145	0.1376
HT-MNPO (Armo) vs RONPO (2457)	647	0.3617	0.1051
RONPO (1400) vs RONPO (2457)	647	0.4645	0.3385

Table 13: Selected GPT-5.5 pairwise judge results. “Left WR” is the win rate of the first method in the pair; values below 0.5 favor RONPO checkpoint 2457 in the first four rows.